

Simplex Method

Motivation:

The **algebraic method** of solving Linear Programming Problems works by enumerating all basic feasible solutions (BFS). Although simple in principle, this approach is highly impractical:

- For a problem with n variables and m constraints, the number of possible bases is $\binom{n}{m}$, which grows combinatorially.
- For example, with $n=20$ and $m=10$, there are more than 184,756 possible basic solutions to check.

Simplex method overcomes this computational difficulty:

- It moves systematically from one BFS (vertex) to another adjacent BFS, always improving (or maintaining) the objective function.
- It exploits the geometry of convex polytopes: since the optimal solution lies at a vertex, only vertices need to be examined.
- In practice, the simplex method visits only a very small fraction of all possible BFS before reaching the optimum.

In short: The algebraic method is useful for understanding the theory of LPP, but the simplex method is essential for efficient computation when the number of variables and constraints is large.

Ex 1: Solve the following LPP by simplex method:

$$\text{Max } Z = 2x_1 + 3x_2$$
$$\text{Subject to: } 2x_1 + x_2 \leq 4, \quad x_1 + 2x_2 \leq 5, \quad x_1 \geq 0, x_2 \geq 0.$$

Solution: Let us introduce the slack variables s_1 and s_2 in the first and second constraints, respectively.

$$\text{Max } Z = 2x_1 + 3x_2 + 0.s_1 + 0.s_2$$
$$\text{Subject to: } 2x_1 + x_2 + s_1 = 4$$
$$x_1 + 2x_2 + s_2 = 5, \quad x_1, x_2, s_1, s_2 \geq 0.$$

		$C_j \rightarrow$	2	3	0	0	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	
0	s_1	4	2	1	1	0	4/1
0	s_2	5	1	2	0	1	5/2 $\rightarrow$
$Z_j - C_j \rightarrow$		$Z = 0$	-2	-3 $\uparrow$	0	0	
0	s_1	3/2	3/2	0	1	-1/2	1 $\rightarrow$
3	x_2	5/2	1/2	1	0	1/2	5
$Z_j - C_j \rightarrow$		$Z = 7.5$	-1/2 $\uparrow$	0	0	3/2	
2	x_1	1	1	0	2/3	-1/3	
3	x_2	2	0	1	-1/3	2/3	
$Z_j - C_j \rightarrow$		$Z = 8$	0	0	1/3	4/3	

Since, $Z_j - C_j \geq 0, \forall j$.
The optimality condition hold.
The optimal solution is
 $x_1 = 1, x_2 = 2$
 $Z_{max} = 8.$

Ex 1: Solve the following LPP by simplex method:

$Max\ Z = 6x_1 + 5x_2$

Subject to: $x_1 + x_2 \leq 5, \ 3x_1 + 2x_2 \leq 12, \ x_1 \geq 0, x_2 \geq 0.$

Solution: Let us introduce the slack variables s_1 and s_2 in the first and second constraints, respectively.

$Max\ Z = 6x_1 + 5x_2 + 0.s_1 + 0.s_2$

Subject to: $x_1 + x_2 + s_1 = 5$

$3x_1 + 2x_2 + s_2 = 12, \ x_1, x_2, s_1, s_2 \geq 0.$

		$C_j \rightarrow$	6	5	0	0	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	
0	s_1	5	1	1	1	0	5
0	s_2	12	3	2	0	1	4 $\rightarrow$
$Z_j - C_j \rightarrow$		$Z = 0$	-6 $\uparrow$	-5	0	0	
0	s_1	1	0	1/3	1	-1/3	3 $\rightarrow$
6	x_1	4	1	2/3	0	1/3	6
$Z_j - C_j \rightarrow$		$Z = 24$	0	-1 $\uparrow$	0	2	
5	x_2	3	0	1	3	-1	
6	x_1	2	1	0	-2	1	
$Z_j - C_j \rightarrow$		$Z = 27$	0	0	3	1	

Since, $Z_j - C_j \geq 0, \forall j.$
The optimality condition hold.

The optimal solution is
 $x_1 = 2, x_2 = 3$
 $Z_{max} = 27.$

Ex 1: Solve the following LPP by simplex method:

$$\begin{aligned} \text{Min } Z &= x_1 - 3x_2 + 2x_3 \\ \text{Subject to: } 3x_1 - x_2 + 2x_3 &\leq 7, \quad -2x_1 + 4x_2 \leq 12, \quad -4x_1 + 3x_2 + 8x_3 \leq 10, \quad x_i \geq 0. \end{aligned}$$

Remark: Since this is a minimization problem, we can proceed in two ways:

Convert the problem into maximization type, using $\min f(x) = -\max f(x)$.

Or,

Proceed with the same minimization problem with the criteria of the entering variable corresponds to the maximum values of $Z_j - C_j$ and the optimality criteria $Z_j - C_j \leq 0, \forall j$.

Solution: Let us introduce the slack variables s_1, s_2 and s_3 in the first, second and third constraints, respectively.

$$\text{Max } Z = -x_1 + 3x_2 - 2x_3 + 0.s_1 + 0.s_2 + 0.s_3$$

$$\text{Subject to: } 3x_1 - x_2 + 2x_3 + s_1 = 7$$

$$-2x_1 + 4x_2 + s_2 = 12,$$

$$-4x_1 + 3x_2 + 8x_3 + s_3 = 10, \quad x_1, x_2, s_1, s_2 \geq 0.$$

		$C_j \rightarrow$	-1	3	-2	0	0	0	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	x_3	s_1	s_2	s_3	
0	s_1	7	3	-1	2	1	0	0	—
0	s_2	12	-2	4	0	0	1	0	3 $\rightarrow$
0	s_3	10	-4	3	8	0	0	1	10/3
$Z_j - C_j \rightarrow$		$Z = 0$	1	-3 $\uparrow$	2	0	0	0	
0	s_1	10	5/2	0	2	1	1/4	0	4 $\rightarrow$
3	x_2	3	-1/2	1	0	0	1/4	0	—
0	s_3	1	-5/2	0	8	0	-3/4	1	—
$Z_j - C_j \rightarrow$		$Z = 9$	-1/2 $\uparrow$	0	2	0	3/4	0	
-1	x_1	4	1	0	4/5	2/5	1/10	0	
3	x_2	5	0	1	2/5	1/5	3/10	0	
0	s_3	11	0	0	10	1	-1/2	1	
$Z_j - C_j \rightarrow$		$Z = 11$	0	0	12/5	1/5	8/10	0	

Since, $Z_j - C_j \geq 0, \forall j$.

The optimality condition hold.

The optimal solution is

$$x_1 = 4, x_2 = 5, x_3 = 0$$

$$Z_{max} = 11 \Rightarrow Z_{min} = -Z_{max} = -11$$

Exercises

Solve the following LPP by simplex method:

1.
$$\text{Max } Z = 4x_1 + 5x_2$$

Subject to: $2x_1 + 3x_2 \leq 12,$
 $2x_1 + x_2 \leq 4, \quad x_1 \geq 0, \quad x_2 \geq 0.$
2.
$$\text{Max } Z = 3x_1 + 2x_2 + 4x_3$$

Subject to: $x_1 + x_2 + x_3 \leq 6,$
 $2x_1 + x_2 + 3x_3 \leq 10$
 $x_1 + 4x_2 + 2x_3 \leq 12, \quad x_1 \geq 0, \quad x_2 \geq 0.$

Big-M Method

The **simplex method** works smoothly when the problem can be written in **standard form with only slack variables** (i.e., all constraints are " $\leq$ ")

But in practice, linear programming problems often have constraints of type:

- " $\geq$ " (greater than or equal to), or
- " $=$ " (equal to).

In such cases, introducing slack variables is **not enough** to reach a basic feasible solution (BFS).

For example, with a " $\geq$ " constraint, we introduce a **surplus variable** (subtracted), which destroys feasibility. To restore feasibility, we must introduce an **artificial variable**.

Now the challenge is:

- Artificial variables are **not real decision variables**; they are just a device to start the simplex method.
- We must ensure that in the **optimal solution**, these artificial variables **vanish (become 0)**.

This is exactly where the **Big-M method** helps:

- Assign a **very large penalty ($-M$ for maximization)** to each artificial variable in the objective function.
- This forces the simplex algorithm to drive artificial variables out of the basis (since keeping them would reduce the objective value drastically).
- Thus, the method allows us to **apply simplex systematically** even when no obvious BFS exists at the start.

Ex 1: Solve the following LPP by Big-M method:

$$\begin{aligned} \text{Max } Z &= 5x_1 + 4x_2 \\ \text{Subject to: } x_1 + x_2 &\geq 5, \\ 2x_1 + x_2 &= 6, \quad x_1, x_2 \geq 0. \end{aligned}$$

Solution: Let us introduce the surplus variables s_1 in the first constraint and artificial variables A_1, A_2 in the first and second constraints, respectively. Also consider the cost component $M(>> 0)$ for the artificial variables A_1 and A_2 .

Therefore, the standard form of the given LPP is given by:

$$\begin{aligned} \text{Max } Z &= 5x_1 + 4x_2 + 0.s_1 - M.A_1 - M.A_2 \\ \text{Subject to: } x_1 + x_2 - s_1 + A_1 &= 5 \\ 2x_1 + x_2 + A_2 &= 6, \quad x_1, x_2, s_1, A_1, A_2 \geq 0, \end{aligned}$$

		$C_j \rightarrow$	5	4	0	$-M$	$-M$	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	A_1	A_2	
$-M$	A_1	5	1	1	-1	1	0	5
$-M$	A_2	6	2	1	0	0	1	3 $\rightarrow$
$Z_j - C_j \rightarrow$			$-3M - 5 \uparrow$	$-2M - 3$	M	0	0	
$-M$	A_1	2	0	1/2	-1	1	$-1/2$	4 $\rightarrow$
5	x_1	3	1	1/2	0	0	1/2	6
$Z_j - C_j \rightarrow$			0	$-M/2 \uparrow$	M	0	$3M/2 + 1/2$	
4	x_2	4	0	1	-2	2	-1	—
5	x_1	1	1	0	1	-1	1	$\rightarrow$
$Z_j - C_j \rightarrow$		$Z = 21$	0	0	$-3 \uparrow$	$M + 3$	$M + 1$	
4	x_2	6	2	1	0	0	1	
0	s_1	1	1	0	1	-1	1	
$Z_j - C_j \rightarrow$		$Z = 24$	3	0	0	M	$M + 4$	

Since, $Z_j - C_j \geq 0, \forall j$. Hence, the optimality condition hold and the optimal solution is

$$x_1 = 0, x_2 = 4, s_1 = 1, A_1 = A_2 = 0$$

$$Z_{max} = 24.$$

Big-M optimality & infeasibility tests

Let $Z_j - C_j$ denote the reduced cost of column j :

Optimality: If $Z_j - C_j \geq 0 \forall j$ (for maximizing problem) and in the associated basic solution *every artificial variable has value 0* (whether non-basic, or basic with zero level), then the current basic solution is an *optimal solution* of the original LPP.

Infeasibility: If $Z_j - C_j \geq 0 \forall j$, but *at least one artificial variable is basic with strictly positive value*, then the original LP is *infeasible* (no feasible solution exists).

Ex 1: Solve the following LPP:

$$\begin{aligned} \text{Max } Z &= 2x_1 + 5x_2 \\ \text{Subject to: } 3x_1 + 2x_2 &\geq 6, \\ 2x_1 + x_2 &\leq 2, \quad x_1, x_2 \geq 0. \end{aligned}$$

Solution: Let us introduce a surplus variables s_1 and an artificial variable A_1 in the first constraints, and a slack variable in the second constraint. Also consider the cost component $M(>> 0)$ for the artificial variables A_1 .

Therefore, the standard form of the given LPP is given by:

$$\begin{aligned} \text{Max } Z &= 2x_1 + 5x_2 + 0.s_1 + 0.s_2 - M.A_1 \\ \text{Subject to: } 3x_1 + 2x_2 - s_1 + A_1 &= 6 \\ 2x_1 + x_2 + s_2 &= 2, \quad x_1, x_2, s_1, A_1, s_2 \geq 0, \end{aligned}$$

		$C_j \rightarrow$	2	5	0	0	$-M$	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	A_1	
$-M$	A_1	6	3	2	-1	0	1	2
0	s_2	2	2	1	0	1	0	1 $\rightarrow$
$Z_j - C_j \rightarrow$			$-3M - 2 \uparrow$	$-2M - 5$	M	0	0	
$-M$	A_1	3	0	$1/2$	-1	$-3/2$	1	6
2	x_1	1	1	$1/2$	0	$1/2$	0	2 $\rightarrow$
$Z_j - C_j \rightarrow$			0	$-\frac{M}{2} - 4 \uparrow$	M	$\frac{3M}{2} + 1$	0	
$-M$	A_1	2	-1	0	-1	-2	1	—
5	x_2	2	2	1	0	1	0	$\rightarrow$
$Z_j - C_j \rightarrow$		$Z = 21$	$M + 10$	0	M	$2M + 5$	0	

Here, we can see $Z_j - C_j \geq 0 \forall j$ but the artificial variable A_1 is present in the final table with positive value $A_1 = 2$. So, the given LPP has infeasible solution.

Case: Unbounded Solution

Ex: Solve the following LPP by Big-M method:

$$\begin{aligned} \text{Max } Z &= 3x_1 + 2x_2 + x_3 \\ \text{Subject to: } 2x_1 + x_2 + x_3 &= 12, \\ 3x_1 + 4x_3 &= 11, \quad x_1 \in \mathbb{R}, x_2, x_3 \geq 0. \end{aligned}$$

Solution: Since x_1 is unrestricted in sign, we substitute $x_1 = x'_1 - x''_1$ in the given LPP

$$\begin{aligned} \text{Max } Z &= 3x'_1 - 3x''_1 + 2x_2 + x_3 \\ \text{Subject to: } 2x'_1 - 2x''_1 + x_2 + x_3 &= 12, \\ 3x'_1 - 3x''_1 + 4x_3 &= 11, \quad x'_1, x''_1, x_2, x_3 \geq 0. \end{aligned}$$

Introduce the artificial variable A_1 in the second constraint.

$$\begin{aligned} \text{Max } Z &= 3x'_1 - 3x''_1 + 2x_2 + x_3 - M.A_1 \\ \text{Subject to: } 2x'_1 - 2x''_1 + x_2 + x_3 &= 12, \\ 3x'_1 - 3x''_1 + 4x_3 + A_1 &= 11, \quad x'_1, x''_1, x_2, x_3, A_1, A_2 \geq 0. \end{aligned}$$

		$C_j \rightarrow$	3	-3	2	1	-M	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x'_1	x''_2	x_2	x_3	A_1	
2	x_2	12	2	-2	1	1	0	12
-M	A_1	11	3	-3	0	4	1	11/4 $\rightarrow$
$Z_j - C_j \rightarrow$			$-3M+1$	$3M-1$	0	$-4M+3$	0	
2	x_2	37/4	5/4	-5/4	1	0	-1/4	-
1	x_3	11/4	3/4	-3/4	0	1	1/4	-
$Z_j - C_j \rightarrow$			1/4	-1/4	0	0	M-1/4	

The final table describes, the variable x''_2 want to enter the table, but none of the basic variable is leaving as $a_{12}, a_{22} < 0$.

Hence, the given LPP has **unbounded solution**.